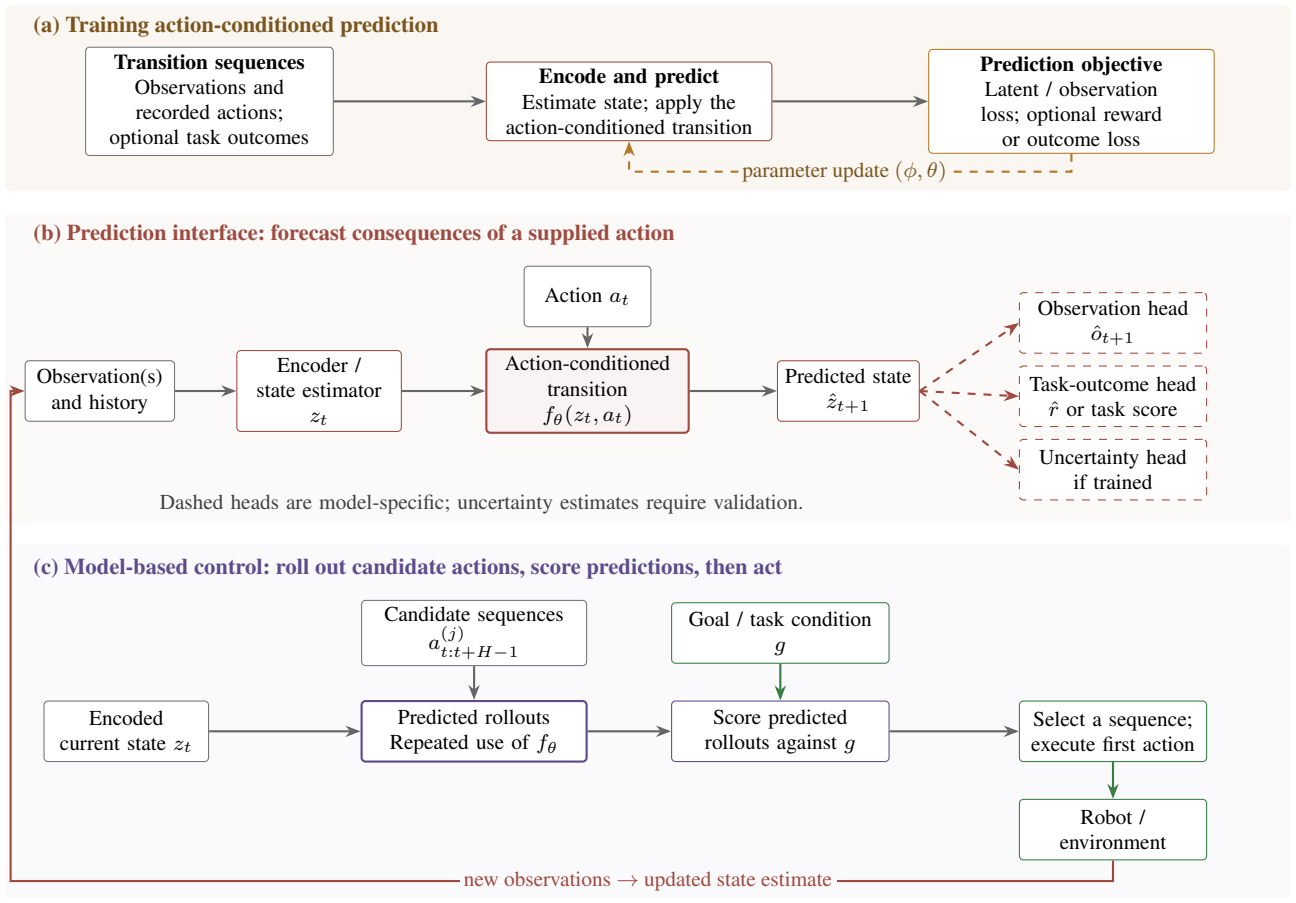

(a) Training action-conditioned prediction
Transition sequences
Observations and recorded actions; optional task outcomes
Encode and predict
Estimate state; apply the action-conditioned transition
Prediction objective
Latent / observation loss; optional reward or outcome loss
parameter update (ϕ, θ)
(b) Prediction interface: forecast consequences of a supplied action
Action a_t
Observation(s) and history
Encoder / state estimator z_t
Action-conditioned transition $f_\theta(z_t, a_t)$
Predicted state $\hat{z}_{t+1}$
Observation head $\hat{o}_{t+1}$
Task-outcome head $\hat{r}$ or task score
Uncertainty head if trained
Dashed heads are model-specific; uncertainty estimates require validation.
(c) Model-based control: roll out candidate actions, score predictions, then act
Candidate sequences $a^{(j)}_{t:t+H-1}$
Goal / task condition g
Encoded current state z_t
Predicted rollouts Repeated use of f_θ
Score predicted rollouts against g
Select a sequence; execute first action
Robot / environment
new observations → updated state estimate